

Zachary Mineiko

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EDUCATION

University of Iowa, College of Engineering

B.S. in Electrical Engineering, Computer Science Focus

Coursework: Manufacturing Processes, Linear Systems, Computer Architecture, Comm. Networks, Generative Artificial Intelligence

May 2025

Iowa City, IA

TECHNICAL SKILLS

Engineering & Design Tools: SolidWorks, Autodesk Fusion 360, KiCad, AutoCAD, ModelSim, Atmel Studio

Manufacturing & Process: PCB Design, Root Cause Analysis, Process Automation, Workflow Optimization, Tolerance Analysis

Languages & Data Tools: Python, MATLAB, Bash, OpenCV, Matplotlib, SQL, C/C++

EXPERIENCE

Full Stack Engineer (Contract)

Aug 2025 – Present

FlipWyz

Remote

- Applied analytical and data driven problem solving to identify system bottlenecks and drive measurable improvements in operational performance
- Designed and enforced structured workflows and process standards across a development team to improve consistency and reduce errors

Engineering Technician Intern

Mar 2022 – Aug 2022

Firefly

Chicago, IL

- Identified recurring hardware failures across a deployed product fleet and eliminated root causes through structured corrective action, directly reducing defect recurrence and improving production output quality
- Analyzed repair workflows and redesigned work sequences and procedures, increasing daily unit throughput by 20% and transforming the Chicago branch from lowest to highest performing location
- Developed and standardized repair documentation and procedures to ensure process consistency, reduce variability, and support repeatable results across technicians
- Built data collection and process automation tools using Python and OpenCV to track production metrics and identify efficiency opportunities across the operation
- Contributed to cross-functional coordination through weekly stand-ups, collaborating with engineers and production personnel to surface and resolve operational issues

Electrician's Assistant

May 2020 – May 2023

Custom Craft Builders

Chicago, IL

- Led installation teams for low-voltage electrical systems across commercial job sites, coordinating work sequences and personnel to meet project timelines
- Interpreted engineering drawings and wiring specifications to plan and execute code-compliant installations, identifying producibility risks before work began
- Coordinated with project leads and production personnel across multiple sites to schedule, sequence, and execute installations efficiently

PROJECTS

Smart Pet Feeder – Senior Design Project | *ESP32, Autodesk Fusion 360, SolidWorks, PCB Design, PostgreSQL*

- Led end-to-end development of a multi-sensor IoT system, managing work sequences from hardware design through firmware, physical assembly, and backend integration
- Designed and 3D modeled the full physical enclosure and mechanical assemblies, controlling dimensional tolerances to ensure proper component fit, sensor alignment, and repeatable full-system manufacturability
- Created custom PCB schematics and component footprints in Autodesk Fusion 360, reviewing designs for producibility and integrating multiple sensors into a unified, manufacturable platform
- Reviewed system architecture at each stage for quality and producibility risks, iterating on design to eliminate failure points before final assembly

IR Receiver with Real-Time Signal Processing | *ESP32, Embedded C, Analog & Digital Filtering*

- Designed and validated a hybrid analog/digital detection system from initial circuit layout through final performance verification in high-noise environments
- Reviewed design at each stage for quality and reliability risks, iterating on signal conditioning and filtering to eliminate failure modes and achieve consistent detection up to 42 inches

Process Automation – Firefly Data Pipeline | *Python, OpenCV, Matplotlib*

- Identified manual data entry as a key bottleneck in the repair operation and independently designed and built an automated data collection pipeline to eliminate it
- Deployed production tracking tools that surfaced actionable metrics, enabling data-driven decisions on workflow layout and resource allocation